A Project Plan for Comparing Statistical and Machine Learning Models for Electricity Demand Forecasting in NSW

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1. Introduction and Motivation

The National Electricity Market (NEM) coordinates electricity generation, trading and distribution across eastern and southern Australia, matching electricity supply with demand in near real time [1]. As electricity cannot be stored at scale, the NEM depends on electricity forecasts to ensure that supply and demand are balanced [2]. Accurate forecasting supports Australia's progress toward international sustainability commitments, such as the United Nations Sustainable Development Goal 7, an area in which Australia has been assessed as "off track" [3]. Additionally, this dependence on accurate forecasting carries financial incentives. For example, generators may rely on these forecasts to decide whether to keep their plants running for further demand or shut down to save costs, however risking the added expense of restarting if demand rises unexpectedly [4].

Because of these real financial consequences and broader sustainability commitments, organisations seek demand analysis to make informed planning and investment decisions. This project is based on a brief from Endgame Economics, a consultancy which provides this kind of analysis. Their clients are typically market bodies that rely on accurate demand forecasts to inform decisions such as policy and investments [5]. Therefore, in this project we propose the following research question; "Which forecasting models perform best for medium to long term electricity demand in New South Wales when evaluated across seasonal variation, temperature changes, and differing demand levels?"

Answering this question directly supports the kind of decision making our potential client may rely on, helping market bodies identify which modelling approaches are most robust under v arying conditions, and in turn informing more confident policy and investment decisions. To answer this question, we intend to apply a combination of statistical and machine learning techniques, evaluating each against the seasonal, temperature, and demand level conditions. By using statistical methods, we intend to capture the seasonality of electricity demand. While using machine learning methods, we intend to capture the non-linear, temperature driven variability in demand that time series models alone may not fully capture. By the end of this project, we aim to deliver a comparative evaluation of these models, leveraging existing literature and identifying which best balances accuracy and interpretability across NSW's seasonal and temperature conditions.

2. Brief Literature Review

National Energy Retail Law (NSW) requires companies to provide a reliable supply of energy; achieve targets set by the Australian Energy Market Commission (AEMC); and reduce Australia's greenhouse gases [6]. To ensure sustainability and reliability, efficient and cost-effective prediction of energy demand is imperative for energy retailers [7, 8]. Previous research suggests four possible approaches to meet the needs of retailers considering dynamic city-level data; Time Series Models; Regression Models; Convolutional Neural Networks; and Hybrid Models.

Daily load half hour readings from the Transmission Company of Nigeria (Aug-Dec, 2021) accurately predicted by LTSM, indicating successful short-term prediction, with similar time-step parameters to the current situation [14, A4]. Likewise, in comparison to XGB and ARIMA, LTSM was most accurate; predicting South African energy consumption [13, A3]. Applied to Shanghai office building energy consumption data (24th June- 15 Sept, 2013), LTSM outpaced Backpropagation Network (BP); SVM; and ANN[15, A5]. Further LTSM accuracy obtained via optimisation for timestep [15, A6]. Comparison of U.S. energy consumption trends revealed LR marginally outperformed RF and XGB [11]. XGB trailed RF and LR on Accuracy; precision exposed LR's outperformance [11, A2]. Demonstrating importance of tuning hyperparameters, optimized RF outperformed all compared models (ARIMA; ETS; Prophet; MLP; SVM; ANFIS; GRNN; LTSM; RandNN; XGB, LightGBM; FNM, Nadaraya-Watson Estimator; MTGNN and hybrid ES-adRNNe) [12]. Renewable energy focused city-level dynamic datasets (Vaasa, Kalmar, Brunei, Thailand) included similar features: Barometric pressure, temperature, humidity, sun radiation, holidays to make short-term load predictions with 1hr and 30min timestep data [10, 16-18]. Vaasa study revealed

XGBoost (XGB) the most reliable model over Multiple Linear Regression (MLR); while in Brunei, comparison between ARIMA, SARIMA and hybrid Holt-Winters+SARIMA gave SARIMA the highest recommendation [10, 16, A1, A7]. To forecast Thailand's city region, hybrid CLTSM-RNN produced fortified predictive power against LSTM; RNN and CNN [18]. Hybrid models reveal requirement to balance accuracy with computational cost [16]. In Kalmar, XGB obtained highest error in training, while SARIMAX performed consistently mid-range for both training and testing [17, A8, A9]. FBP outperformed SARIMAX in testing; however, all models produced lower predictive performance than CNN for both training and testing [17]. Sun radiation contributes to prediction accuracy due to consumer solar panels [11, 16, 17]. Holidays apply pressure on demand [17].

Prediction Accuracy is associated with granularity of data [14,17,18]. Given current dataset has a high granularity of 30 min timesteps, modelling choice pursues balance between prediction accuracy and computational cost; providing energy companies with efficient and economically sound forecasts [6-7, 16]. Due to suburb-scope of Bankstown, NSW, there is no spatial variation to consider; therefore, it is suggested to focus on time series and regression methods, reducing resource requirements [12, 15, 17, 18]. Alternatively, exploratory data analysis may investigate increased timesteps to reduce model load and determine impact on accuracy [15]. A leading limitation within literature appears to be the absence of temperature fluctuations consistent with Australian specific-location standards. [A10, 19-25]. The current temperature dataset obtained from Bankstown, NSW contains a much higher temperature range (Max: 44.7°C; Min:-1.3°C; Range: 46°C), presenting an avenue to explore the prediction capabilities of SARIMA, Prophet, Random Forest, XGBoost and LightGBM within broader fluctuation bounds. Due to the nature of energy demand, research question focus and consistent incorporation of multivariate datasets across the research, it is recommended to obtain further data such as school holidays, humidity, sun radiation and/or cloud cover corresponding to decreased solar panel capability and therefore higher demand [10, 11].

3. Methods, Software and Data Description

The project will use a combination of statistical and machine learning methods to forecast electricity demand in New South Wales. This approach reflects findings in the literature that no single modelling technique is sufficient for capturing both the regular seasonal structure of demand and the strong temperature driven variability observed in the NEM [1,7]. The modelling strategy begins with simple, well-established benchmarks and then evaluates more flexible models against them, ensuring that the final choices are supported by evidence rather than complexity alone.

This project will model half-hourly New South Wales electricity demand using a combined statistical and machine learning approach to capture both regular seasonal structures and extreme temperature driven variability [1,7]. To systematically improve upon baseline operator forecasts [1], classic time series methods will be evaluated against flexible machine learning algorithms. SARIMA will handle daily and weekly cycles, while Prophet will model overarching trends and holiday effects with temperature as an exogenous variable [7]. To overcome traditional time series limitations in capturing non-linear weather interactions, tree-based models, specifically Random Forest, XGBoost, and LightGBM, will be applied using lagged demand, calendar features, and weather inputs [7,9–12,17]. Neural networks such as LSTM and support vector machines will remain secondary options due to excessive tuning overhead for structured tabular data [13–15]. Additionally, synthetic demand generation will be incorporated to model full distribution scenarios, directly addressing the reliance on single point forecasts identified in existing literature [7,8]. These modelling choices directly support the research question by enabling performance comparisons across seasonal variation, temperature changes and differing demand levels.

loModels will be assessed using standard error metrics, including mean absolute error, root mean squared error and mean absolute percentage error, consistent with established practice in the forecasting literature [7,9–12]. Because the data are time-ordered, training will occur on earlier periods and testing on later periods, with a separate validation set used for tuning to avoid leakage. As we will be comparing multiple models, we will use consistent features, test and train sets, alongside standardised model evaluation (MSE, MAE and MAPE).

The quantitative workflow will rely primarily on Python libraries including pandas, NumPy, statsmodels, pmdarima, Prophet, scikit-learn, XGBoost, and LightGBM to maintain reproducible modelling pipelines [5,7]. TensorFlow and Keras will be reserved solely for potential LSTM implementations, while R tools such as ggplot2 and R Markdown will provide visualisation and reporting capabilities [5].

The project uses three NSW datasets from 2010 onwards, sourced from National Electricity Market and meteorological archives [1]. The primary target variable will be half-hourly electricity demand, paired with irregular temperature readings from the Bankstown weather station and historical operator forecasts. Data preparation will require standardising timestamps, resampling irregular temperature data to a matching half-hourly grid, imputing short gaps, and engineering calendar and lagged demand features.

This methodology assumes that seasonal patterns and temperature effects remain stable enough over the medium term to support model training, and that Bankstown temperature readings reasonably represent conditions across the NSW region. It also assumes that resampling irregular temperature data preserves the underlying weather signal. Finally, it assumes that operator forecasts included in the dataset reflect realistic baseline performance, allowing meaningful comparison between statistical and machine-learning approaches under varying seasonal and demand conditions.

4. Activities and Schedule

Group A meets three times weekly: a Thursday consultation with Wei Tian, a team stand-up immediately after, and a Sunday working session. Extra meetings occur before submissions. Swapnil Shinde, elected team leader in Week 0, chairs all sessions; agendas are posted to GitHub 24 hours before consultations and minutes within 24 hours after. The 33 activities sit on a Miro board maintained by Xavier Sun, each with a phase, owner, effort estimate, priority and assessment link. Code merges to the main branch only through reviewed pull requests, and an activity is complete once merged, rerun end-to-end from cleaned data, and written into the report.

The activities fall into six phases (Appendix B), each feeding the next. Work is front-loaded into data preparation because medium- to long-term forecasting depends more on design than fitting. Recent demand lags do not exist months ahead, so features are limited to calendar terms, harmonic seasonality, long-run trend and temperature. Temperature is also unknown at long horizons, so each model is fitted twice: once with observed temperature and once with analogue-year temperature, with the gap treated as a result. Modelling begins once the feature set and rolling-origin splits are finalised on 12 September. The six models run in parallel, meet a shared evaluation harness on 16 September, and freeze on 20 September ahead of the Week 4 checkpoint.

Appendix C table outlines each member's role. The structure is symmetric: every member owns one data or exploratory task, one forecasting model, and one report chapter. On 3 September the group agreed that each member should build and validate a model end-to-end so all can speak directly to the modelling process and avoid bottlenecks. Whoever drafted a section of this plan owns the corresponding report chapter, ensuring reading done now is not duplicated in Week 6. The board allocates 14–20 days of work per member across seven weeks, sized against availability recorded in Week 0.

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6. Appendix

Appendix A — Reported Model Performance Metrics

Table A1. Electricity demand, Vaasa — short-term hourly [10]

Model	MSE	RMSE	MAE	MAPE	R^2
XGBoost	21.22	4.61	3.61	0.063	0.902
Multiple Linear Regression	21.40	4.626	3.66	0.066	0.901
Random Forest	23.19	4.82	3.67	0.061	0.892

Table A2. U.S. energy consumption trends — classification metrics [11]

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.701	0.70	0.70	0.70
Random Forest	0.696	0.70	0.70	0.70
XGBoost	0.67	0.67	0.67	0.67

Table A3. South African energy consumption [13]

Model	MAE
LSTM	0.2
XGBoost	—
ARIMA	—

Table A4. Transmission Company of Nigeria, Aug–Dec 2021 — half-hourly load [14]

Model	MAPE	RMSE
LSTM	0.01	19.759

Table A5. Shanghai office building, 24 Jun – 15 Sep 2013 [15]

Model	MAPE
LSTM	9%
Backpropagation Network (BP)	12%
Artificial Neural Network (ANN)	16%
Support Vector Machine (SVM)	21%

Table A6. LSTM timestep optimisation, Shanghai office building [15]

Timestep	MAPE

6 h	3%
18 h	5%
24 h	6%

Table A7. Active power usage, Brunei — short-term [16]

Model	RMSE
ARIMA	0.048
SARIMA	0.6201
Holt-Winters + SARIMA (hybrid)	1.15

Table A8. City energy consumption, Kalmar, Sweden — training [17]

Model	MSE	RMSE	MAE	MAPE	R^2
Random Forest	64.61	8.03	6.19	4.67%	66%
XGBoost	119.82	10.94	8.63	6.51%	38%
SARIMAX	91.29	9.55	3.71	2.75%	52%
FB Prophet	—	—	—	—	—
CNN	13.36	8.21	2.75	0.02%	0.93
ARIMAX	—	—	—	—	—

Table A9. City energy consumption, Kalmar, Sweden — testing [17]

Model	MSE	RMSE	MAE	MAPE	R^2
Random Forest	278.19	16.67	14.43	9.35%	64%
XGBoost	—	—	—	—	—
SARIMAX	133.72	11.56	9.87	6.28%	20%
FB Prophet	97.43	9.87	7.95	5.22%	0.42
CNN	67.60	8.22	5.74	4.35%	0.60
ARIMAX	—	—	—	—	—

Dataset: hourly datapoints 2021–2023, features include temperature and price; FB Prophet additionally models holidays.

Table A10. Temperature Ranges (Sweden; USA; South Africa; Nigeria; Brunei; Shanghai) [19-25]

Location	Temperature Range
Sweden	~29°C

USA	~22°C
South Africa	~32°C
Nigeria	~32°C
Brunei	~10°C
Shanghai	~33°C
Thailand	~22°C

Appendix B — Phases, activities and outputs

Phase	**Activities** *(Task ID)*	**Est. Date Range**	**Planned Output**
Project Planning	Task 1 to 8; Task 33	31/08/2026 ~ 08/09/2026	Working contract, agreed research question, the project plan and each member's checklist
Data Engineering & Preparation	Task 10 to 11	07/09/2026 ~ 12/09/2026	Cleaned half-hourly dataset, horizon-safe feature set, rolling - origin splits
Exploratory Data Analysis	Task 12 to 13	09/09/2026 ~ 12/09/2026	Load profiles, the temperature-demand response curve, trend and structural breaks
Modelling	Task 14 to 19	14/09/2026 ~ 15/09/2026	Six forecasters built in parallel, one per member, on a common feature set
Evaluation & Validation	Task 20 to 20	16/09/2026 ~ 23/09/2026	Stratified error tables, a frozen model set, one improvement round on the winner
Reporting & Communication	Task 9; Task 23 to 32	21/09/2026 ~ 13/10/2026	Weekly reflections; the final 35-page group report and 15-minutes video.

Appendix C — Roles and justification

Member	Role	Assignment Basis
Swapnil Shinde	Team Lead	Elected in Week 0, and holds the broadest skill profile in the group: Python, R, SQL, statistics, visualisation and reporting. Chairs the stand-up and the lecturer consultation, arbitrates where the group is split, and holds the statistical stream, the temperature–demand response (Task 13) followed directly by SARIMAX (Task 15). Writes up software, data and pre-processing (Task 26).
Daniel Cao	Machine Learning Engineer	ML engineering and data science experience, and custodian of the group repository. Owns the ingest-and-clean pipeline (Task 10) and the Random Forest (Task 16), then writes the data cleaning, assumptions and modelling methods (Task 27) documenting work done firsthand rather than inherited.
Kelly Hamilton	Data Science Research	A psychology background trained in research design and literature synthesis, with statistics and machine learning on top. Owns the literature review and the discussion (Task 25), the two chapters that frame the project and then interpret its results, and builds the LSTM (Task 19) so that the deep learning comparison is argued by the member writing the discussion.
Faiza Mumtaz	Data Scientist	ML engineering combined with data analysis, Tableau and reporting. Produces the load-profile and seasonality analysis (Task 12) and the XGBoost model (Task 17), then writes the Exploratory Data Analysis chapter (Task 28) built from that same analysis.
Arman Hajisafi	Data Engineer	PySpark and data engineering experience, the direct match for the horizon-safe feature set and the rolling-origin split design (Task 11) over eleven years of half-hourly data, the single activity every later result depends on. Also builds LightGBM (Task 18), pairing with XGBoost for a like-for-like comparison, and writes the introduction, conclusion and reference organisation (Task 24).
Xavier Sun	Product	Works as a product manager for data science and ML engineering teams, so owns the board, schedule and risk log (Task 33), and keeps the plan current as scope moves. A commerce and accounting background anchors the translation of model output into the business framing the non-technical needs. Builds the baseline models (Task 14) that every other result is measured against, writes the analysis and results chapter (Task 29), and produces the video (Task 32).